TERRASHIFT AI — BUSINESS MODEL CANVAS

STRATEGIC ENTERPRISE SPECIFICATION · VISION 2030 & GLOBAL EARTH OBSERVATION

KEY PARTNERS #7

- **Space Agencies:** ESA Copernicus Sentinel-2 open constellation & NASA USGS Landsat.
- **Commercial Tasking:** Planet Labs (3m daily SuperDoves) & Maxar (30cm high-res optical).
- **STAC & Cloud Catalogs:** Element84 & AWS Open Data Earth Search cloudless composites.
- **Sovereign Hosting:** KSA cloud zones (AWS Bahrain/GCP Dammam/CNTXT) for data residency.
- **GIS Ecosystem:** Esri ArcGIS Enterprise and QGIS plugin interoperability networks.
- **Academic R&D;:** KAUST & KACST for marine blue-carbon & arid soil flux verification.

KEY ACTIVITIES #5

- **Siamese AI R&D;:** Training bi-temporal ResU-Net models on LEVIR-CD & OSCD benchmarks.
- **Automated Ingest:** Continuous Sentinel-2 orbital scene ingestion & disk tile caching.
- **Predictive BFAST:** Harmonic breakpoint decomposition & 2030 completion trajectories.
- **Carbon Modeling:** IPCC Tier-1 4-pool biomass flux & thermal ΔLST urban cooling.
- **Audit Generation:** Producing tamper-evident verifiable PDF compliance dossiers.

KEY RESOURCES #6

- **Proprietary Checkpoints:** Trained weights (siamese_unet_checkpoint.pt) with τ calibration.
- **Global Tile Archives:** Multi-year Sentinel-2 cloudless composite cache (2017–2025).
- **Compute Infrastructure:** High-throughput NVIDIA GPU clusters for low-latency SSE inference.
- **Interdisciplinary Team:** Remote sensing geodesists, computer vision ML scientists & full-stack eng.
- **Regional Baselines:** Curated ground-truth masks across NEOM, Riyadh, Red Sea & Amazon.

VALUE PROPOSITIONS #1

- **Zero-Noise Change AI:** Physics-guided Siamese U-Net neutralizes solar angle and seasonal noise down to exact m².
- **BFAST 2030 Trajectory:** Non-parametric breakpoint forecasting isolates structural velocity and completion horizons.
- **IPCC Carbon Verification:** 4-pool carbon accounting (AGB, BGB, SOC, Dead Wood) + urban heat island mitigation.
- **Verifiable Audit Dossiers:** Instant one-click executive PDFs embedding raw satellite rasters and STAC scene IDs.
- **95% Cost & 100x Speedup:** Replaces $1,200/km² drone and manual surveyor passes with automated <10s satellite telemetry.
- **Cross-Module Continuity:** Custom user-drawn AOI instantly synchronized across Map, Timeline, Benchmarks & Carbon.

CUSTOMER RELATIONSHIPS #4

- **Solutions Engineering:** Dedicated GIS architects assisting with historical baseline & CRS setup.
- **Automated Monitoring:** Scheduled 5-day Sentinel-2 pass ingestion with threshold breach alerts.
- **Audit-Grade Deliverables:** Cryptographically referenced compliance PDFs ready for board / ministry review.
- **Enterprise SLA:** 99.9% API uptime, dedicated GPU inference queues, and model retraining cycles.

CHANNELS #3

- **Direct Enterprise / B2G:** Dedicated sales to Vision 2030 Delivery Units, PIF entities & master EPCs.
- **Web Map Studio:** Self-service browser SaaS application with interactive WebGL layers & loupe.
- **REST & STAC 1.0.0 API:** Headless machine-to-machine endpoints for enterprise data lakes & GIS.
- **GIS Marketplace Plugins:** Native extensions for Esri ArcGIS Pro & QGIS workflows.
- **Industry Summits:** Live keynotes & telemetry demonstrations at LEAP, Cityscape & COP Summits.

CUSTOMER SEGMENTS #2

- **Megaproject PMOs & EPCs:** NEOM (The Line/Oxagon), Red Sea Global, ROSHN, Qiddiya, Diriyah Gate.
- **Sovereign Urban Planners:** Royal Commission for Riyadh City (RCRC), MEWA, Saudi Green Initiative (SGI).
- **ESG & Carbon Auditors:** Independent carbon offset registries, Verra, RVCM & climate compliance bodies.
- **Infrastructure Insurers:** Global reinsurance firms & sovereign wealth funds auditing asset deployment.
- **Multi-Sided Dynamics:** Developers track execution speed; regulators & investors independently verify claims.

COST STRUCTURE #9

- **Cloud Compute & GPU Inference (25–30%):** NVIDIA A100/L40S clusters, high-speed NVMe tile caches, vector storage.
- **Commercial Satellite Ingestion (20–25%):** Wholesale API licensing for on-demand PlanetScope 3m & Maxar 30cm tasking passes.
- **R&D; & Engineering Payroll (35–40%):** Specialized compensation for Computer Vision Researchers, Geodesists & Full-Stack Engineers.
- **Enterprise Sales & Compliance (10–15%):** Government RFP procurement teams, ISO/SOC2 audits, sovereign cloud certifications.

REVENUE STREAMS #8

- **Annual SaaS Subscriptions:** Pro ($24k/yr, 2.5k km²), Enterprise ($75k/yr, 15k km²), Sovereign/Gov ($250k+/yr, unlimited).
- **Usage-Based AOI Volume Surcharge:** Metered expansion pricing at $0.75 – $2.50 per monitored km² / month beyond base tier.
- **High-Res Tasking Pass-Through (25% Margin):** Direct on-demand commercial satellite tasking markup processed via Siamese engine.
- **Bespoke Enterprise Deployments:** Custom on-premises air-gapped sovereign installations & custom land-class model calibration.